Weekly Progress Report

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I. Overview:

During Week 3 of the internship, the focus was on implementing the core banking functionalities of the Banking Information System using Core Java. Building upon the registration and login modules developed in the previous week, this phase involved developing features such as deposit, withdrawal, and balance checking. Additionally, emphasis was placed on implementing basic error handling to ensure reliable and user-friendly operations. The system was tested through a console-based interface to validate correct functionality and data handling.

II. Achievements:

1. Deposit Functionality:

The deposit feature was successfully implemented, allowing users to add money to their account. The system updates the account balance immediately after a successful transaction.

- Users can enter the deposit amount.
- The system validates the input amount.
- The balance is updated dynamically.

2. Withdrawal Functionality:

The withdrawal feature was developed to allow users to withdraw money from their account. Proper validation ensures that users cannot withdraw more than their available balance.

- Users enter the withdrawal amount.
- The system checks for sufficient balance.
- The balance is updated after a successful transaction.

3. Balance Checking:

A balance inquiry feature was implemented to allow users to view their current account balance at any time.

4. Error Handling Implementation:

Basic error handling mechanisms were introduced to improve system reliability.

Handled cases include:

- Invalid input amounts (zero or negative values)
- Insufficient balance during withdrawal

5. Menu-Based User Interface Enhancement:

A user-friendly menu was introduced after login, allowing users to navigate between different banking operations easily.

Example:

1. Deposit
2. Withdraw
3. Check Balance
4. Logout

III. Challenges:

1. Implementing Transaction Validation:

Ensuring that invalid inputs such as negative or zero values were handled correctly was a challenge.

Solution:

Conditional checks were added before processing transactions to validate the input amount.

2. Handling Insufficient Balance:

Implementing logic to prevent withdrawal beyond available balance required careful condition handling.

Solution:

A comparison check between withdrawal amount and account balance was implemented.

IV. Lessons Learned:

During this week, several important concepts were learned and applied:

- Implementation of real-world banking operations using Core Java
- Use of conditional statements for validation and error handling
- Enhancement of menu-driven console applications
- Improved understanding of object-oriented programming concepts

V. Next Week's Goals:

The objectives for Week 4 include completing the remaining features of the Banking Information System:

- Fund Transfer
- Transfer money between accounts
- Account Statement
- Display transaction history
- Advanced Error Handling
- Improve validation for better system stability
- Final Project Integration
- Combine all features into a complete system

VI. Additional Comments

Week 3 marked a significant step in transforming the project into a functional banking system. The successful implementation of deposit, withdrawal, and balance checking features enhanced the usability and practicality of the application. The project is progressing well towards completion, with only a few advanced features remaining for the final week